

Lab	Practical
LAB-1	Revision of HTML & CSS. - Write HTML, CSS3 code for following animation effects: (A) - changing background color of div from red to yellow. - moving div from Left to right, right to bottom, bottom to left and left to top. - put delay of 2s in animation. - Change animation-iteration count property of program 2 to run infinite times. - Change animation direction property to run it in reverse order. - Change value of animation-timing-function property to change the speed of animation. - Write an HTML snippet with CSS for following effects: (A) - Link after visited red color before visited orange color. - Add special style (change color, font size etc..) to first line of paragraph. - Add special style (change color, font size etc..) to first letter of paragraph. - Change background color to yellow when user select particular text. - Add special style to an element when user mouse over it. - Design complete personal website using CSS. (A)
LAB-2	Revision of Bootstrap. - Write HTML, Bootstrap code for following effects: (A) - Use Bootstrap typography as per below image. - Use Bootstrap typography to make text look like below image. - Use Bootstrap typography to put abbreviation in paragraph. - Use Bootstrap typography to align text (left, center, right). - Create Unstyled list using Bootstrap typography. - Use Bootstrap to display responsive image in webpage. Also use alignment to align it in left, center and right side of webpage. - Use Bootstrap to create thumbnail of image. - Create Static website of 4-5 pages using HTML, CSS and Bootstrap. (A)
LAB-3	Revision of Bootstrap. Desing clone of darshan university home page using bootstrap. (A)

LAB-4	Introduction of JavaScript and string functions. 1. Write a program to demonstrate internal javascript. (A) 2. Write a program to demonstrate external javascript. (A) 3. Write a program to concat two or more strings. (A) 4. Write a program to demonstrate the use of various inbuilt string functions like charAt(), charCodeAt(), startsWith(), endsWith(), includes(), indexOf(), lastIndexOf(), match(), repeat(), replace("", ""), search(), split(), substr(start,length), substring(start, end), toString(), trim(). (A) 5. Write a program to explore more inbuilt string functions like slice(), toLowerCase(), toUpperCase(), valueOf(), fromCharCode(). (A)
LAB-5	Working with number handling functions, operators and if condition. 1. Write a program to demonstrate number handling functions of javascript. (A) 2. Write a program to demonstrate the use of arithmetic operators using if condition. (A) 3. Write a program to create calculator using switch case. (A) 4. Write a program to display result of a student whether he/she pass or fail the exam. (Criteria is >39 marks is all 5 subjects.) Also display the grade of the student. (Criteria >90 then grade 'A', >80 then 'B', >70 then 'C', >60 then 'D', >40 then 'E', <40 then 'F'). (B) 5. Write a program to swap values of 2 variables without using 3rd variable. (B)
LAB-6	Implementation of if condition inside loops. 1. Write a JavaScript program to print the factors of a given number. (A) 2. Write a JavaScript program to display all prime numbers between a given range of numbers. (A) 3. Write a JavaScript program to print the GCD (Greatest Common Divisor) of two numbers. (A) 4. Write a JavaScript program to take a 2-digit number and separate its digits. Then, multiply the first digit by itself as many times as indicated by the second digit. (B) For example, if the number is 23, it should be separated into 2 and 3. Then, 2 should be multiplied by itself 3 times ($2 \times 2 \times 2 = 8$). 5. Write an HTML page with JavaScript that takes a number from a popup box in the range 0–999 and displays it in words. If the number is out of range, it should display "Out of Range", and if the input is not a number, it should display "Not a Number" in the result box. (A) 6. Write an HTML file with JavaScript that finds the position of the first occurrence of the vowel "a" and the last occurrence of the vowel "a" in a given word, and returns the string between them. For example, for the word "ajanta", the script should return: • First occurrence of "a": Position 1 • Last occurrence of "a": Position 6 • String between them: "jant". (B)

LAB-7	Pattern Printing Programs Using JavaScript. <table><tr><td><pre>* ** *** **** *****</pre></td><td><pre>***** **** *** ** *</pre></td><td><pre> * ** *** **** *****</pre></td><td><pre> * ** *** **** *****</pre></td></tr><tr><td>(A)</td><td>(A)</td><td>(A)</td><td>(A)</td></tr></table> <table><tr><td><pre> * ** *** ** *</pre></td><td><pre>1 11 121 1331 14641</pre></td><td><pre>1 2 2 3 3 3 2 2 1</pre></td><td><pre> * *** ***** ***** *****</pre></td></tr><tr><td>(B)</td><td>(B)</td><td>(B)</td><td>(B)</td></tr></table>	<pre>* ** *** **** *****</pre>	<pre>***** **** *** ** *</pre>	<pre> * ** *** **** *****</pre>	<pre> * ** *** **** *****</pre>	(A)	(A)	(A)	(A)	<pre> * ** *** ** *</pre>	<pre>1 11 121 1331 14641</pre>	<pre>1 2 2 3 3 3 2 2 1</pre>	<pre> * *** ***** ***** *****</pre>	(B)	(B)	(B)	(B)
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LAB-8	JS UDF, PopUp boxes and validations. 1. Write a JavaScript program to create a simple calculator using popup boxes. (A) 2. Write a JavaScript program to check whether a given number is prime or not using a popup box. (A) 3. Write a JavaScript program to find the factorial of a given number using a popup box. (A) 4. Write a JavaScript program to print the Fibonacci series up to a given number using a popup box.(A) 5. Write a JavaScript program to check whether a given number is a palindrome or not using a popup box. (A) 6. Write a JavaScript program to check whether a given number is an Armstrong number or not using a popup box. (A) 7. Write a JavaScript program to demonstrate all three types of popup boxes. (B) 8. Write a program using HTML, CSS, and JavaScript to design a simple calculator with a GUI (Graphical User Interface). (C)																

LAB-9	Implementing validation in forms and pages. - Write a JavaScript program to validate user data given from the HTML form: (A) - Username must contain a minimum of 8 characters. - Password must contain at least one digit and one special character, and its length should be between 8 and 12 characters. - Password and Confirm Password must be the same. - Age must be greater than 18 years (calculate from Date of Birth). - Enrollment number must contain exactly 11 digits. - Validate the email format. - Write a program to prepare a student registration form and validate it using JavaScript (Name, Email, Contact Number). (A) - Write a program to prepare an employee form and validate it using JavaScript (Email, Contact Number, Gender, City, Username, Password, Confirm Password). (A) - Write a program to create a static login page using HTML and JavaScript. (B) - Write a program to create a scientific graphical calculator using HTML and JavaScript. (B)
LAB-10	Working with DOM. - Write a program to demonstrate the use of the getElementById() method in JavaScript. (A) - Write a program to demonstrate the use of the getElementsByName() method in JavaScript. (A) - Write a program to demonstrate the use of the getElementsByTagName() method in JavaScript. (A) - Write a program to demonstrate the use of DHTML (Dynamic HTML). (A) - Write a program to print the current date and time when the "Date and Time" button is clicked. (B)
LAB-11	DOM Manipulation Using JavaScript. - Create a webpage that changes text content when a button is clicked. (A) - Create a page that changes the background color of the webpage using a button. (A) - Create a program to show and hide a paragraph using a toggle button. (A) - Take input from a user (name) and display a greeting message using DOM. (A) - Create a program to change the image when a button is clicked. (A) - Create a counter application that increases value on button click. (A) - Create a list where new items are added using a button. (B) - Create a program to remove the last item from a list. (B)

LAB-12	Implementing mouse event. 1. Write a program to change the background color when a button is clicked. (A) 2. Write a program for the following effect: If the mouse is moved over the heading, change the font color to “red”, and if the mouse moves out of the heading, change it back to “black”. (A) 3. Write a program to recognize which mouse event is fired. (A) 4. Write a JavaScript program that handles the following mouse events. Add the necessary HTML elements: (B) a. If the mouse is moved over the heading, its color should change to yellow, and when the mouse moves out, it should change to black. b. If the “Find Time” button is clicked, display the current date and time. c. If the button named “Red” is clicked, the background color should change to red. d. If the button named “Green” is clicked, the background color should change to green. 5. Write a JavaScript program that handles the following keyboard events. Add the necessary HTML elements: (B) a. Display the key code of the key pressed. b. If the key pressed is “a”, “e”, “i”, “o”, or “u”, the script should display that a vowel key is pressed. c. When the key is released, the background color should change to blue. 6. Write a program using HTML and JavaScript that asks the user to enter their date of birth. On clicking the “Calculate” button, the program should calculate how many days are left until the user’s next birthday. (C)
LAB-13	Implementing keyboard event. 1. Write a program to greet the user according to the current time For example, if a user opens the page at 08:00 AM, the program should display “Good Morning”. (A) 2. Write a program to recognize which keyboard event is fired. (A) 3. Write a program to demonstrate changing the properties of an HTML element using JavaScript. (B) 4. Write a program to take a character input from the user and display whether the key pressed is a number, an alphabet, or a symbol. (C)
LAB-14	Understanding the new concepts of ES-6. 1. Write a program to demonstrate the concept of classes in JavaScript. (A) 2. Demonstrate the difference between let, var, and const in JavaScript. (A) 3. Write a program to demonstrate the concept of arrow functions in JavaScript. (A) 4. Write a program to demonstrate arrow functions with only one argument in JavaScript. (A) 5. Write a program to demonstrate callbacks in JavaScript. (A)

LAB-15	Understanding the new concepts of ES-6. 1. Write a program to demonstrate arrow functions without arguments in JavaScript. (A) 2. Write a program to check whether a given number is prime or not using an arrow function in JavaScript. (A) 3. Write a program to check whether a given number is a palindrome or not using an arrow function in JavaScript. (B) 4. Write a program to demonstrate multiple arrow functions in JavaScript. (B) 5. Write a program to print prime numbers between two given numbers using an arrow function in JavaScript. (C)
LAB-16	Working with ES-6 array methods. 1. Write a program to demonstrate the use of various array methods such as join() and split() in JavaScript. (A) 2. Write a program to demonstrate the use of various array methods such as includes() and find() in JavaScript. (A) 3. Write a program to demonstrate the features/functions of ES6 in JavaScript. (B) 4. Write a program to append an object to an array using JavaScript. (B)
LAB-17	Working with ES-6 array methods. 1. Write a program to demonstrate some function of ES6. (A) 2. Write a program to demonstrate the filter() function of ES6. (A) 3. Write a program to demonstrate the reduce() function of ES6. (A) 4. Write a program to demonstrate the map() function of ES6. (A) 5. Write a program to create an array containing color names in lowercase letters (e.g., ['red', 'green', 'blue']) and convert each array element into capitalized form using the map() method. (B) 6. Write a program to perform insert, update, and delete operations on a static array using JavaScript. (C)

LAB-18	Implementation of array methods of ES-6 in programs. 1. Write a program to create a numerical array containing number elements. Use the filter() method to return numbers greater than 50. (A) 2. Write a program to create an array of student names and print each element using the ES6 forEach() method. (A) 3. Write a program to create a numerical array and remove duplicate elements from the array using the reduce() method. (B) 4. Write a program to perform CRUD (Create, Read, Update, Delete) operations on a static array using JavaScript. (B) 5. Write a JavaScript program that handles the following mouse events: (B) a. If the left mouse button is pressed in the browser, display the message "Left Clicked". b. If the right mouse button is pressed in the browser, display the message "Right Clicked". 6. Write a JavaScript program containing a list of checkboxes. When a checkbox is selected, display the selected values in comma-separated format. (For example, use roll numbers as checkbox values and display the selected roll numbers in comma-separated format.). (C) 7. Create a Snake game using JavaScript. (C)
LAB-19	Implementation of Destructuring and Working with import, export and spread operators. 1. Write a program to demonstrate the use of object destructuring in JavaScript. (A) 2. Write a program to demonstrate the use of array destructuring in JavaScript. (A) 3. Write a program to demonstrate the use of the export module in JavaScript. (A) 4. Write a program to demonstrate the use of the import module in JavaScript. (A) 5. Write a program to demonstrate the spread operator in JavaScript. (B) 6. Write a program to combine two numerical arrays using the spread operator in JavaScript. (C)
LAB-20	Demonstration of ReactJS. 1. Setting up react environment. (A) 2. Hello world webapp using ReactJS. (A) 3. Demonstrate the use of JSX in ReactJS. (A) 4. Write a program to create a simple class component in ReactJS. (A) 5. Write a program to create a simple functional component in ReactJS. (A)
LAB-21	Implementation of basic programs of ReactJS. 1. Create a function component in separate file and link with App.js. (A) 2. Create a class component in a separate file and link it with App.js. (A) 3. Demonstrate the use of props in ReactJS. (A) 4. Demonstrate event handling in ReactJS. (A) 5. Write a ReactJS program to display an element only if its attribute is Display is set to true (using conditional rendering). (B)

LAB-22	Implementation of basic programs of ReactJS. 1. Demonstrate the use of the map() method in ReactJS to display array elements. (A) 2. Display faculty data stored in an array using ReactJS. (B) 3. Display student data stored in an array using ReactJS. (B) 4. Display product data stored in an array using ReactJS. (B) 5. Display faculty, student, and product data stored in arrays using ReactJS with navigation. (C)
LAB-23	Demonstration of Routing in ReactJS. 1. Implementing Routing in ReactJS. (A) 2. Develop basic website using 5 different component (pages) and implement Routing in it. (i.e. About, Contact etc...). (B) 3. Build a React component with form fields (like name, email, etc.), capture the input using state, and display the entered data on the same or another component. (B)
LAB-24	Implementation of Routing in ReactJS. 1. Create an Employee form in ReactJS to accept user input and display it. (A) 2. Apply CSS styles to React components using traditional CSS files, inline styles, and CSS modules. (B) 3. Demonstrate styling with conditional classes or styles. (B)
LAB-25	Demonstration of hooks in ReactJS. 1. Demonstrate useState hook in ReactJS. (A) 2. Demonstrate useEffect hook in ReactJS. (A) 3. Write a program to create a simple calculator using ReactJS. (A) 4. Print "Hello world" after every 5 seconds also have a state named count which updates when we click on the button. (B) 5. Write a program to perform CRUD operation on array using ReactJS. (B)
LAB-26	Demonstration of MockAPI. 1. Create a MockAPI online with following fields: (FacultyID, FacultyName, FacultyExp, FacultyImage). (A) 2. Write a program to Perform a POST Request in React With Fetch API. (A) 3. Write a program to Perform a DELETE Request in React With Fetch API. (A) 4. Perform CRUD operation on MockAPI using ReactJS. (B)
LAB-27	Demonstration of CRUD operation using MockAPI. 1. Demonstrate CRUD operation using REST API. (A) 2. Perform CRUD operation on multiple API (minimum 3) with routing for each API. (B) 3. Perform CRUD operation on multiple API (minimum 10) with routing for each API. (C)

LAB-28	Implementation of CRUD operation using MockAPI. 1. Implement search functionality on API data using ReactJS. (A) 2. Create a dynamic form using ReactJS with add/remove fields. (B) 3. Create a table component to display API data with sorting and filtering. (C)
LAB-29	Perform Static Authentication and manage the state across layout using useContext. 1. Create an app using ReactJS with login functionality and use the current user state throughout the application using useContext hook. (A) 2. Create a mini project using mockAPI. (B)
LAB-30	Mini Project. 1. Create a mini project using mockAPI. (A)